

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	K.Bhavana	Reg. No.:	192524178
Date:	27-07-2026	Duration:	60 minutes

Code of Conduct

I K. Bhavana (192524178) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Bhavana_____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

- 1. Understand the Problem Statement**
 - Explain the problem in your own words.
 - Identify the objectives of the proposed system.
 - State the scope of the problem.
- 2. Identify Key Stakeholders and Challenges**
 - Identify all the stakeholders involved in the system.
 - Explain the roles and responsibilities of each stakeholder.
 - Discuss the major challenges and issues faced by the stakeholders.
- 3. Analyze the Current Approach**
 - Describe the existing system or current process.
 - Identify its strengths and limitations.
 - Analyze the problems or gaps that need to be addressed.
- 4. Propose a Suitable Solution Using Software Engineering Principles**
 - Design a software-based solution for the given problem.
 - Describe the major functional modules of the proposed system.
 - Identify the functional and non-functional requirements.
 - Explain how software engineering principles such as modularity, scalability,

maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Table of Contents

Table of Contents	2
1. Introduction	4
1.1. Background of the Problem	4
1.2. Importance of Solving the Problem	4
1.3. Related SDG Goal(s)	4
1.4. Objectives of the Case Study	4
2. Problem Understanding	5
2.1. Problem Description	5
2.2. Objectives of the Proposed Software	5
2.3. Scope	5
In Scope	5
Out of Scope	5
3. Stakeholder Analysis	5
4. Existing System Analysis	7
4.1. Existing Process	7
4.2. Strengths	7
4.3. Weaknesses	7
4.4. Gap Analysis	7
5. Proposed Software Solution	8
5.1. Proposed System	8
5.2. Functional Modules	8
5.3. Functional Requirements	8
5.4. Non-Functional Requirements	9
5.5. Software Engineering Principles Applied	9
6. System Design	11
6.1. Use Case Diagram	11
6.2. Use Case Description (Sample)	11
6.3. Activity / Workflow Diagram	11
6.4. System Architecture Diagram	12
6.5. Database ER Diagram	13
6.6. Wireframe / UI Screens	15
7. Methodology / SDLC Selection	17
7.1. Selected Model: Agile (Scrum Framework)	17
7.2. Why This Model?	17
7.3. Advantages	17

7.4.Suitability	17
7.5.Development Stages (Sprints)	17
8. Implementation Plan	18
9. Expected Outcomes	19
9.1.Benefits	19
9.2.Business / Administrative Impact	19
9.3.User Benefits	19
9.4.Social Impact and SDG Contribution	19
10. Risks and Limitations	20
11. Conclusion	21
12. References	22
13. Appendix	22
13.1.Sample Survey Questions (for Entrepreneur Feedback)	22
13.2.Additional Notes	22

1. Introduction

1. Background of the Problem

Women entrepreneurs play an increasingly vital role in economic development, yet a large number of women-owned small and micro businesses struggle to grow due to limited access to government schemes, financial assistance, business training, and professional mentoring. Information about applicable schemes, subsidies, loan programs, and skill-development courses is currently scattered across multiple government department websites, banking portals, and NGO platforms, each with its own registration process, eligibility criteria, and application format. This fragmentation forces women entrepreneurs — many of whom are already balancing limited time and resources — to navigate a confusing, inconsistent landscape just to discover what support is even available to them, let alone apply for it successfully.

2. Importance of Solving the Problem

Reducing the friction between women entrepreneurs and the support ecosystem directly increases the number of women-led businesses that survive and scale, which in turn contributes to household income, local employment, and broader economic growth. A single, integrated digital platform can consolidate scheme discovery, funding applications, training enrollment, and mentor/peer networking into one trusted destination, saving time, reducing dropout at every stage of the support journey, and giving government agencies accurate, centralized data on scheme utilization and impact.

3. Related SDG Goal(s)

This case study is aligned with Sustainable Development Goal 5 – Gender Equality, specifically SDG Target 5.5 (ensuring women's full participation in economic life) and SDG Target 5.a (giving women equal rights to economic resources, financial services, and ownership). The proposed portal also supports SDG 8 (Decent Work and Economic Growth) by fostering entrepreneurship and job creation, and SDG 9 (Industry, Innovation and Infrastructure) through the use of a digital platform to strengthen access to institutional support.

4. Objectives of the Case Study

1. To analyze the limitations of the current fragmented approach to supporting women entrepreneurs.
2. To identify the stakeholders involved in the women entrepreneurship support ecosystem.
3. To design a centralized, software-based portal that integrates scheme discovery, funding, training, and mentorship.
4. To define the functional and non-functional requirements of the proposed platform.
5. To recommend a suitable Software Development Life Cycle (SDLC) model for the project.
6. To evaluate the expected outcomes, benefits, risks, and future scope of the proposed solution.

2. Problem Understanding

2.1. Problem Description

Women who own or wish to start small businesses currently face a scattered digital landscape when seeking help: government scheme information is published across multiple ministry and state department websites, loan and subsidy details are held separately on bank or NBFC portals, business skill-training programs are advertised through yet another set of NGO or academic institution websites, and mentorship opportunities — where they exist — are arranged informally through personal networks or one-off events. Because there is no single, authoritative place to discover, compare, and apply for these opportunities, many eligible women either never learn about a scheme that could help them, abandon an application midway due to confusing paperwork, or miss training and networking opportunities altogether. A government agency has therefore proposed developing a unified digital platform to bring registration, funding discovery, training enrollment, and networking together in one place.

2.2. Objectives of the Proposed Software

- Reduce the manual, fragmented effort involved in discovering and applying for government schemes.
- Improve efficiency of the application and approval process for funding and subsidies.
- Increase transparency around scheme eligibility, application status, and disbursement.
- Enhance user experience for women entrepreneurs through a single, easy-to-navigate portal.
- Enable structured business mentoring and peer-to-peer networking opportunities.
- Provide government agencies with consolidated data on scheme reach, usage, and impact.
- Support inclusive economic growth by lowering the barrier to accessing institutional support.

2.3. Scope

In

Scope

- Online registration and business profile creation for women entrepreneurs.
- A searchable, filterable directory of government schemes, grants, subsidies, and loan programs.
- End-to-end online application, document upload, and status tracking for funding schemes.
- A catalog of business training and skill-development programs with online enrollment.
- A mentor-matching and networking module connecting entrepreneurs with mentors and peers.
- Notifications for application status updates, upcoming trainings, and new scheme launches.
- An administrative dashboard for government officials to manage schemes and monitor analytics.
- Role-based access for entrepreneurs, mentors, training providers, and administrators.

Out of Scope

- Direct disbursement of loan amounts (handled by partner banks/financial institutions through integration, not by the portal itself).
- Legal or tax advisory services beyond providing informational resources.
- Physical logistics of organizing offline training venues (handled by respective training providers).

3. Stakeholder Analysis

Successful implementation of the Women Entrepreneurship Support Portal depends on clearly identifying every stakeholder group, their role in the process, and their specific responsibilities. The table below summarizes the key stakeholders involved.

Stakeholder	Role	Responsibilities
Administrator	Government Agency Staff	Configures and publishes schemes, manages training/mentor listings, monitors platform-wide analytics and compliance.
Customer (Entrepreneur)	Woman Business Owner / Aspiring Entrepreneur	Registers a business profile, searches and applies for schemes, enrolls in training, connects with mentors.
Government	Policy Maker / Scheme Owner	Defines eligibility criteria and scheme rules, reviews and approves applications, allocates funding budgets.
Staff (Scheme Officer)	Application Reviewer	Verifies submitted documents, evaluates eligibility, approves or rejects funding applications.
Vendor (Bank / NBFC)	Financial Partner	Processes approved fund disbursement, provides loan product details integrated into the portal.
Mentor / Training Provider	Knowledge Partner	Offers mentorship sessions and training programs, tracks participant progress and completion.
Technician / IT Support	System Maintainer	Installs, configures, and maintains hardware/software infrastructure; resolves technical issues and performs updates.

4. Existing System Analysis

4.1.Existing Process

At present, a woman entrepreneur who wants to access government support must independently search across several ministry and state department websites to find schemes relevant to her business category. If she finds a suitable scheme, she typically has to visit a separate portal (or a physical office) to submit an application, often filling in the same personal and business details multiple times across different systems. Loan or subsidy disbursement is coordinated separately with a bank or NBFC, business training programs are discovered mostly through word-of-mouth, social media groups, or NGO outreach events, and mentorship is arranged informally, if at all, through personal or community networks. There is no single system that tracks a woman's overall journey — from registration, to scheme application, to training completion, to business growth outcomes.

4.2.Strengths

- Individual government schemes and portals are often well-established and legally backed.
- Existing banking and financial institution processes for loan disbursement are mature and regulated.
- Community and NGO-driven training/networking events build strong personal trust and rapport.
- Some digital scheme portals already exist, providing a foundation of digitized scheme data to integrate.

4.3.Weaknesses

- Scheme information is scattered across many disconnected websites and departments.
- Entrepreneurs must repeatedly re-enter the same personal/business information on different portals.
- No unified way to track the status of multiple applications across different schemes or banks.
- Training and mentorship opportunities are discovered inconsistently, often missed entirely.
- Government agencies lack a consolidated view of how many women are benefiting and from which schemes.
- Low awareness and digital literacy further compound the difficulty of navigating fragmented systems.

4.4.Gap Analysis

Comparing the current fragmented ecosystem against what is needed to meaningfully support women entrepreneurs reveals several unresolved gaps: there is no single source of truth for scheme eligibility and application status; there is no integrated system connecting funding, training, and mentorship into one journey; government agencies cannot easily measure the real-world impact of their schemes; and entrepreneurs have no easy way to discover new opportunities as they arise. These gaps directly justify the need for a centralized, integrated Women Entrepreneurship Support Portal.

5. Proposed Software Solution

5.1. Proposed System

The proposed solution is a centralized, web and mobile-based Women Entrepreneurship Support Portal that integrates business registration, government scheme discovery, funding applications, training enrollment, and mentorship/networking into a single digital platform. Women entrepreneurs create one business profile that is used across every module — eliminating repeated data entry across multiple government portals. The system intelligently matches entrepreneurs to relevant schemes and training programs based on their profile, allows them to apply and track funding applications end-to-end, and connects them with mentors and peer communities. Government agencies gain a single administrative dashboard to publish and manage schemes, review applications, and monitor real-time analytics on reach and impact across the entire entrepreneurship support ecosystem.

5.2. Functional Modules

- User/Entrepreneur Registration & Business Profile
- Login & Role-Based Authentication
- Scheme Discovery & Eligibility Matching
- Funding Application & Document Upload
- Application Status Tracking
- Training Program Catalog & Enrollment
- Mentor Matching & Mentorship Management
- Networking / Community Forum
- Notifications & Alerts Engine
- Admin Panel (Schemes, Trainings, Reports)

5.3. Functional Requirements

The table below lists the core functional requirements (FRs) that the proposed system must satisfy.

ID	Functional Requirement
FR1	The system shall allow women entrepreneurs to register and create a detailed business profile.
FR2	The system shall allow registered users, mentors, and staff to securely log in using role-based credentials.
FR3	The system shall allow entrepreneurs to search and filter government schemes, grants, and loan programs by eligibility.
FR4	The system shall allow entrepreneurs to submit funding applications with supporting documents online.
FR5	The system shall allow entrepreneurs to track the real-time status of their submitted applications.
FR6	The system shall allow entrepreneurs to browse and enroll in business training and skill-development programs.
FR7	The system shall match entrepreneurs with suitable mentors based on business sector and stated needs.
FR8	The system shall provide a networking/community forum for entrepreneurs to interact with peers and mentors.

FR9	The system shall automatically send notifications for application updates, training schedules, and new schemes.
FR10	The system shall generate scheme utilization, training participation, and impact reports for administrators.

5.4.Non-Functional Requirements

In addition to functional requirements, the system must satisfy the following non-functional requirements (NFRs) to ensure it performs reliably in a real-world government service context.

Category	Requirement Description
Performance	The system shall respond to user requests (e.g., scheme search, application submission) within 3 seconds under normal load.
Security	All personal, business, and financial data shall be encrypted in transit and at rest, with role-based access control.
Scalability	The system shall support scaling from a pilot state-level deployment to a nationwide platform without redesign.
Maintainability	The system shall be built using modular components so that individual modules can be updated independently.
Availability	The system shall maintain at least 99.5% uptime, particularly during scheme application deadlines.
Usability	The interface shall be simple, accessible, and multilingual to support users with varying digital literacy.
Reliability	The system shall ensure no data loss during application submission or document upload, including retry mechanisms.
Portability	The system shall be accessible via web browsers and mobile devices across major operating systems.

5.5.Software Engineering Principles Applied

The proposed design is guided by core software engineering principles to ensure the system is robust, sustainable, and adaptable to the evolving needs of government entrepreneurship programs.

Principle	How It Is Applied
Modularity	The system is divided into independent modules (registration, scheme matching, funding, training, mentorship, networking) that can be developed, tested, and updated separately.
Scalability	A layered, cloud-hosted architecture allows the platform to scale from a single state pilot to a nationwide deployment by adding server capacity without redesigning the application.
Reliability	Transaction-safe application submissions, automated document validation, and retry-safe uploads ensure application data is never lost or duplicated.
Security	Role-based access control, data encryption, and audit logging protect sensitive personal, business, and financial information from unauthorized access.
Maintainability	Clear separation of presentation, business logic, and data layers, along with documented APIs, allows future developers to easily fix bugs or add new schemes/features.

Reusability	Common services such as authentication, notification, and reporting are built as reusable components that can serve multiple modules and future government programs.
Usability	A simple, multilingual, mobile-friendly interface ensures entrepreneurs with varying digital literacy levels can discover and apply for support easily.
Flexibility	Configurable scheme definitions, eligibility rules, and training catalogs allow the system to adapt to new government programs without core redevelopment.

6. System Design

The following diagrams illustrate the structural and behavioral design of the proposed Women Entrepreneurship Support Portal, prepared to support the requirements defined in Section 5.

6.1. Use Case Diagram

The use case diagram identifies the primary actors (Woman Entrepreneur, Mentor, Training Provider, System Administrator, Government Scheme Official, and Bank/Financial Institution) and their interactions with the core functions of the portal.

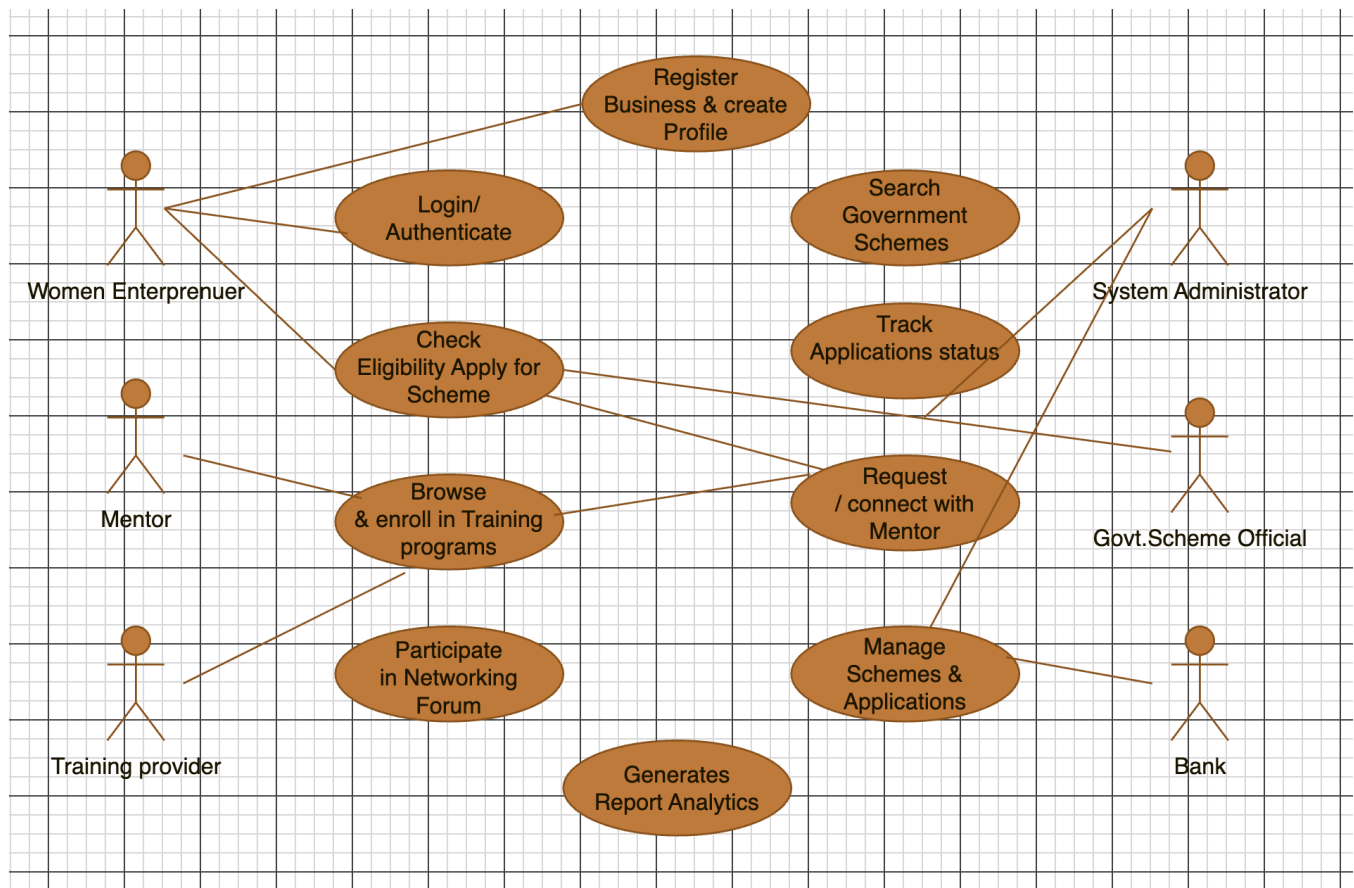


Figure 1: Use Case Diagram – Women Entrepreneurship Support Portal

6.2. Use Case Description (Sample)

Use Case: Apply for Funding Scheme | Primary Actor: Woman Entrepreneur | Precondition: Entrepreneur has a registered business profile. | Main Flow: The entrepreneur searches for schemes matching her business profile, reviews eligibility criteria, and submits an application with supporting documents. The system routes the application to the relevant government official for review, and the entrepreneur is notified of the outcome. | Postcondition: An application record is created and linked to the entrepreneur's profile with a trackable status.

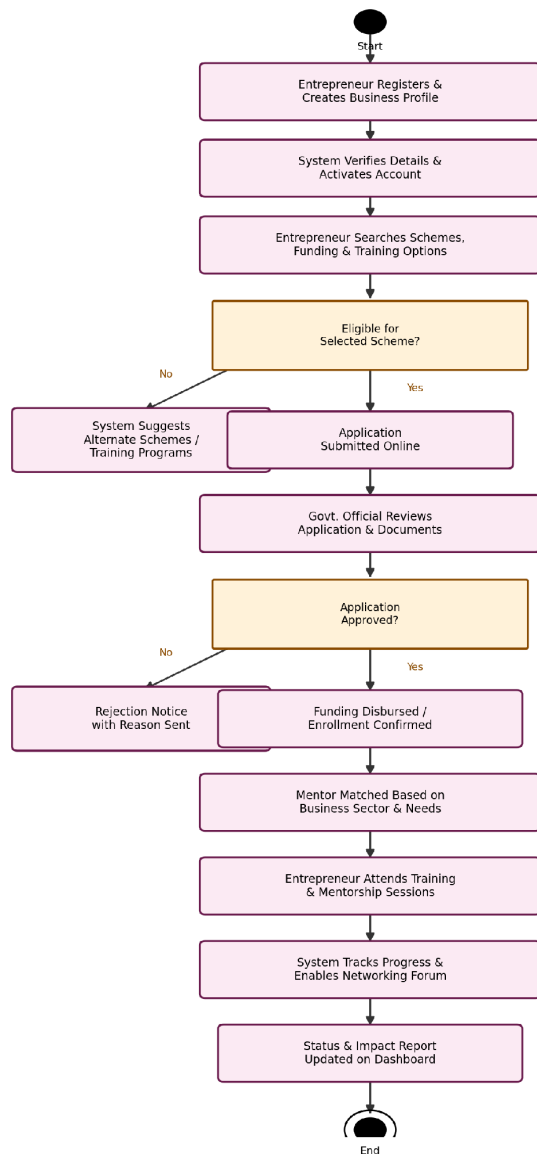


Figure 2: Activity/Workflow Diagram – Scheme Application & Support Process

6.4. System Architecture Diagram

The system follows a layered architecture consisting of a presentation layer (web portal, mobile app, and admin console), an application/business logic layer (registration, scheme matching, training/mentorship, networking and notification services), a data layer (entrepreneur, scheme, and mentor databases), and an external integration layer connecting to government scheme portals, bank/payment gateways, and SMS/email services.

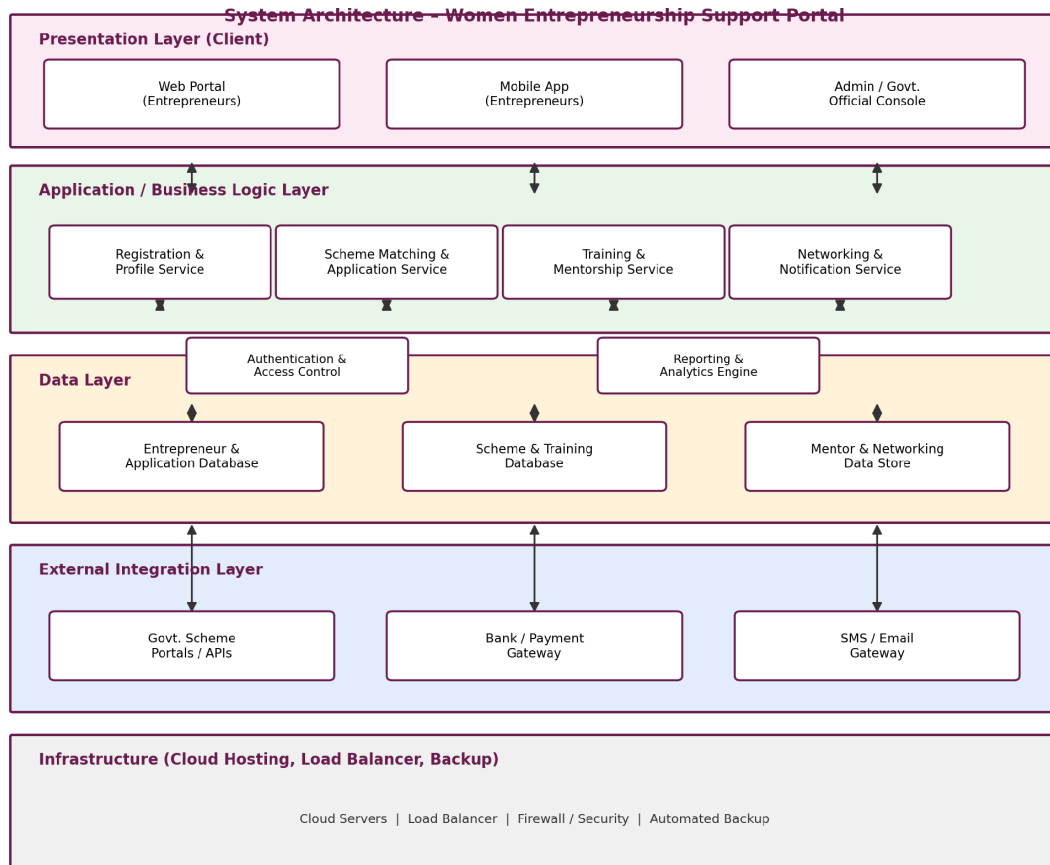


Figure 3: System Architecture Diagram

6.5.Database ER Diagram

The entity-relationship diagram below models the core data entities — Entrepreneur, Scheme, Application, Training Program, Enrollment, and Mentor — along with their relationships and cardinalities.

Entity-Relationship Diagram - Women Entrepreneurship Support Portal

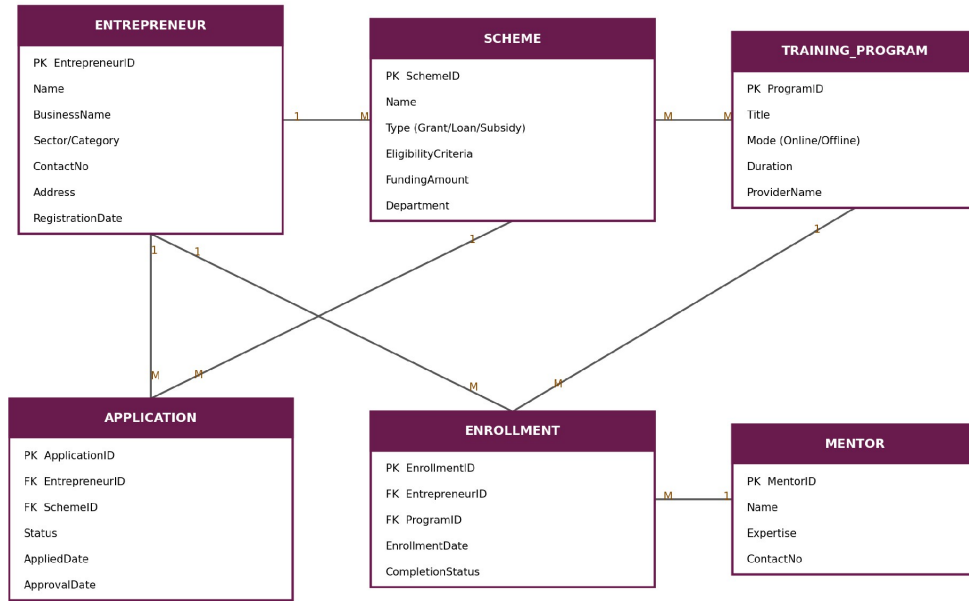
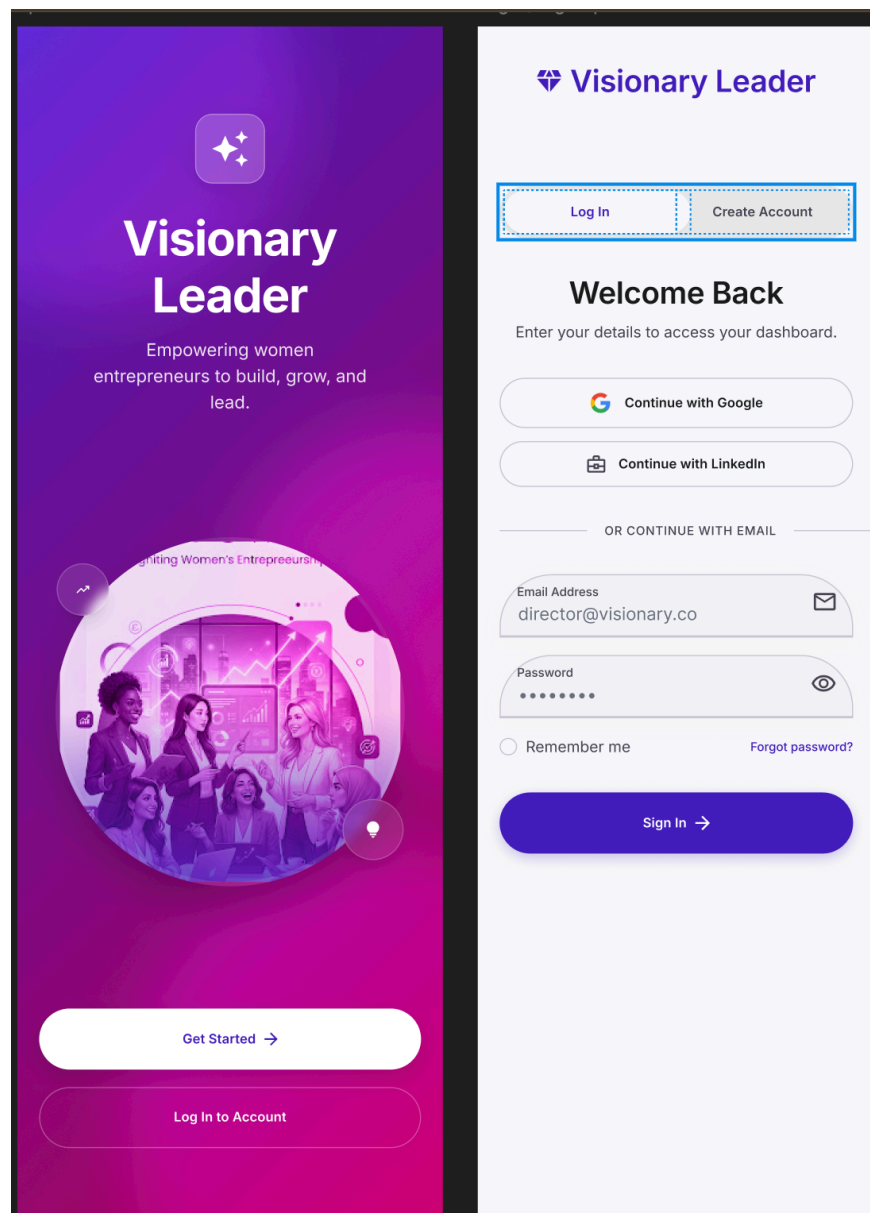


Figure 4: Entity-Relationship (ER) Diagram

6.6.Wireframe / UI Screens

Recommended screens for the entrepreneur and admin-facing applications include: (1) a Registration/ Business Profile screen, (2) a Dashboard showing recommended schemes and trainings, (3) a Scheme Application screen with document upload and status tracker, (4) a Training/Mentor Directory screen with enrollment options, and (5) an Admin Reports screen with scheme utilization charts. It is recommended that students prepare low-fidelity wireframes for these screens using Figma or Canva and attach them in the Appendix.



☰

Visionary Leader

🔔

👋

Hello, Sarah.

Ready to scale your business today?

🔍

Search for schemes, mentors, or resources...

Latest Schemes

View All

Funding

Women Tech Founders Grant

Closes in 12 days

➔

Featured Mentors

See All Mentors



Dr. Elena Rostova
Tech Startup Specialist



Maya Lin
E-commerce Strategist

Quick Links

🔍

My Ideas

💬

Community

📄

Resources

📅

Events

🏠

Home

🏛️

Schemes

📁

Market

👤

Profile

☰

Visionary Leader

🔔

Schemes & Funding

Discover tailored financial support and government initiatives to scale your business.

🔍

Search schemes by name or sector...

All Schemes

Technology

Manufacturing

Central Govt

Eligible

₹

Stand-Up India Scheme

Facilitates bank loans between 10 lakh and 1 crore to at least one Scheduled Caste (SC) or

MAX FUNDING

₹1 Cr

Apply Now ➔

MSME Dept

PMEGP

Prime Minister's Employment Generation Programme aims to generate employment...

SUBSIDY

Up to 35%

Apply Now ➔

State Level

Mudra Yojana for Women

Provides loans to non-corporate, non-farm small/micro enterprises with special...

COLLATERAL

Free

Apply Now ➔

🏠

Home

🏛️

Schemes

📁

Market

👤

Profile

☰

Visionary Leader

🔔

Mentorship

Marketplace

Top Mentors

Connect with experienced leaders in your industry.



Dr. Sarah Jenkins
Tech Startup Advisor

Funding

Tech

➔



Elena Rostova
Retail & E-commerce

Scaling

Retail

➔



Priya Patel
Marketing Strategist

Branding

Social

➔

Masterclass Series

Learn leadership skills from top CEOs.

Explore Classes

🏠

Home

🏛️

Schemes

📁

Market

👤

Profile

☰

Visionary Leader

🔔



Sarah Jenkins

Entrepreneur

📁

My Applications

➔

📖

Saved Schemes

➔

💬

Mentor Chat

➔

🔔

Notifications

➔

⚙️

App Settings

➔

🚪

Logout

Version 2.4.1 (Build 492)

🏠

Home

🏛️

Schemes

📁

Market

👤

Profile

7. Methodology / SDLC Selection

7.1.Selected Model: Agile (Scrum Framework)

The Agile methodology, implemented through the Scrum framework, is recommended for the development of the Women Entrepreneurship Support Portal.

7.2.Why This Model?

Government schemes, eligibility rules, and training programs change frequently as new policies are announced and existing ones are revised. Agile/Scrum allows the development team to deliver working modules (such as registration, scheme discovery, and funding applications) in short, iterative sprints, gather feedback from government stakeholders and pilot users after each sprint, and adjust priorities without disrupting the overall project timeline. This iterative approach reduces the risk of building a platform that does not reflect real scheme rules or genuine entrepreneur needs, which is common with a rigid, single-pass model.

7.3.Advantages

- Incremental delivery allows early modules (e.g., registration and scheme discovery) to be piloted while later modules are still in development.
- Continuous stakeholder feedback from government agencies and entrepreneurs improves requirement accuracy over time.
- Easier to accommodate frequently changing scheme rules, eligibility criteria, or new government programs.
- Regular sprint reviews and demos improve transparency and trust with government stakeholders.
- Reduces overall project risk by identifying usability or workflow issues early through frequent testing.

7.4.Suitability

Given that the portal is government-driven, subject to frequent scheme and policy revisions, and requires close collaboration between developers, government officials, and financial partners, Agile/Scrum is well suited because it embraces change, encourages frequent communication, and delivers usable increments of the platform early — allowing a pilot group of women entrepreneurs to start using core features while the rest of the system is refined.

7.5.Development Stages (Sprints)

7. Sprint 0 – Requirement gathering, stakeholder interviews, and backlog creation.
8. Sprint 1 – Design and development of the Registration and Business Profile module.
9. Sprint 2 – Development of Scheme Discovery and Eligibility Matching module.
10. Sprint 3 – Development of Funding Application, Document Upload, and Status Tracking module.
11. Sprint 4 – Development of Training Catalog, Enrollment, and Mentor Matching modules.
12. Sprint 5 – Development of Networking Forum, Notifications, and Admin Reporting dashboard.
13. Sprint 6 – User acceptance testing (UAT), pilot deployment, and go-live.

8. Implementation Plan

The table below presents a suggested phase-wise timeline for developing and deploying the Women Entrepreneurship Support Portal, aligned with the Agile/Scrum sprints described in Section 7. The modular design also allows additional features and services to be incorporated in the future without affecting the existing functionality of the portal. In the future, the portal can be further enhanced by incorporating advanced technologies such as Artificial Intelligence for personalized scheme recommendations, chatbot-based user support, multilingual interfaces, mobile application support, cloud-based services, digital payment integration, and data analytics for monitoring entrepreneurial growth. These improvements would make the platform more intelligent, accessible, and efficient while meeting the evolving needs of women entrepreneurs.

Phase	Duration
Requirement Gathering & Stakeholder Analysis	1 Week
System Design (UML, Architecture, ER Diagrams, UI Wireframes)	2 Weeks
Development (Sprint-wise module implementation)	6 Weeks
Testing (Unit, Integration, UAT)	2 Weeks
Deployment (Pilot rollout in select districts/states)	1 Week
Training & Documentation for Staff and Administrators	1 Week
Post-deployment Support & Maintenance (Ongoing)	Continuous

9. Expected Outcomes

9.1.Benefits

- Elimination of fragmented, manual searching across multiple government websites.
- Significant reduction in application dropout through a single, guided, end-to-end process.
- Consolidated, accurate data on which schemes, trainings, and mentors are actually being used.

9.2.Business / Administrative Impact

- Government agencies gain real-time visibility into scheme reach and impact, enabling faster policy decisions.
- Reduced administrative overhead in verifying and processing applications manually across departments.
- Better-targeted training and mentorship programs based on actual entrepreneur demand data.

9.3.User Benefits

- Women entrepreneurs discover relevant schemes, funding, and training in one place without repeated paperwork.
- Real-time application status tracking reduces uncertainty and follow-up effort.
- Structured mentorship and networking opportunities help entrepreneurs build skills and business connections.

9.4.Social Impact and SDG Contribution

By simplifying access to funding, training, and mentorship, the platform directly supports increased participation of women in economic activity, advancing SDG 5 (Gender Equality) — particularly the targets on women's equal access to economic resources and financial services. It also contributes to SDG 8 (Decent Work and Economic Growth) by strengthening women-led enterprises, and demonstrates how digital infrastructure can improve the delivery of public institutional support in line with SDG 9 (Industry, Innovation and Infrastructure).

10. Risks and Limitations

- Internet Dependency: Entrepreneurs in rural or remote areas may face unreliable internet connectivity when accessing the portal.
- Data Privacy: The system stores sensitive personal, business, and financial data, requiring strict compliance with data protection regulations.
- Security Threats: A centralized database of applicant and financial data is an attractive target for cyberattacks, requiring robust encryption and access controls.
- Budget Limitations: Government agencies may face funding constraints for full-scale rollout, integration, and staff training.
- User Adoption: Entrepreneurs and staff with low digital literacy may resist transitioning from familiar manual or informal processes.
- Since the system stores personal, business, and financial information of users, there is always a risk of data breaches or unauthorized access if strong security measures are not implemented and maintained.
- The portal may become vulnerable to cybersecurity threats such as hacking, phishing, malware, or ransomware attacks, which can compromise user data and interrupt the normal functioning of the system.
- Developing, deploying, and maintaining the portal requires significant financial investment, and limited budgets may delay implementation, reduce system quality, or restrict future upgrades and enhancements.
- Some women entrepreneurs may have limited digital literacy or lack experience in using online government services, making it difficult for them to register, apply for schemes, or utilize all the features of the portal effectively.
- Integrating the portal with government departments, banks, training organizations, and other external systems may be technically challenging because different organizations use different technologies and data formats.
- As the number of users, schemes, and applications increases, the system must be capable of handling large amounts of data and user traffic without affecting performance or response time.
- Government schemes, eligibility criteria, funding policies, and regulations are updated frequently, requiring regular modifications to the portal to ensure that users always receive accurate and up-to-date information.
- Continuous maintenance is necessary to fix software bugs, improve system performance, update security features, and introduce new functionalities, which requires ongoing technical support and resources.
- The portal depends on users providing accurate personal, business, and financial information, and incorrect or incomplete details may lead to application rejection, processing delays, or incorrect eligibility verification.

- The system must comply with government regulations related to data privacy, cybersecurity, and digital governance, and failure to meet these standards may result in legal issues or loss of public trust.

11. Conclusion

The **Women Entrepreneurship Support Portal** is a comprehensive and user-centric software solution designed to address the major challenges faced by women entrepreneurs in accessing government support services. At present, information related to government schemes, financial assistance, business mentoring, training programs, and networking opportunities is scattered across multiple platforms, making it difficult for entrepreneurs to find relevant information and complete application procedures efficiently. The proposed portal overcomes these challenges by providing a centralized digital platform where all essential services are integrated into a single, secure, and easy-to-use system. The portal simplifies the entire entrepreneurial support process by enabling users to register online, explore government schemes, verify eligibility, apply for funding, upload necessary documents, participate in skill development programs, connect with experienced mentors, and build professional networks. The inclusion of an administrator module ensures efficient management of users, verification of applications, scheme updates, report generation, and overall system monitoring. By automating these processes, the portal minimizes paperwork, reduces manual errors, improves transparency, and significantly decreases the time required for application processing and service delivery. The project adopts the **Agile Software Development Life Cycle (SDLC)**, which is well suited for systems that require continuous improvements and frequent updates. Since government policies, funding schemes, and user requirements change over time, Agile enables developers to deliver the project in small, manageable iterations while incorporating stakeholder feedback at every stage. This iterative approach improves software quality, enhances user satisfaction, reduces development risks, and ensures that the system remains flexible, scalable, and adaptable to future requirements. From a technical perspective, the proposed solution provides a secure, scalable, and maintainable architecture that supports efficient data management and reliable system performance. Proper authentication mechanisms, secure data storage, role-based access control, and regular system maintenance contribute to protecting sensitive user information and ensuring long-term system reliability. The modular design also allows additional features and services to be incorporated in the future without affecting the existing functionality of the portal. In the future, the portal can be further enhanced by incorporating advanced technologies such as Artificial Intelligence for personalized scheme recommendations, chatbot-based user support, multilingual interfaces, mobile application support, cloud-based services, digital payment integration, and data analytics for monitoring entrepreneurial growth. These improvements would make the platform more intelligent, accessible, and efficient while meeting the evolving needs of women entrepreneurs.

12. References

- United Nations. "Sustainable Development Goal 5: Gender Equality." Available at: <https://sdgs.un.org/goals/goal5>
- World Bank. "Women, Business and the Law." Available at: <https://wbl.worldbank.org>
- Sommerville, I. Software Engineering (10th Edition). Pearson Education, 2016.
- Pressman, R. S., & Maxim, B. R. Software Engineering: A Practitioner's Approach (8th Edition). McGraw-Hill Education, 2014.
- IEEE. "IEEE Recommended Practice for Software Requirements Specifications (IEEE 830)." IEEE Standards Association.
- ACM Digital Library. Research articles on Digital Platforms for Entrepreneurship Support. Available at: <https://dl.acm.org>
- Springer. "Digital Government Platforms and Women's Economic Empowerment." Available at: <https://link.springer.com>
- Ministry of Women and Child Development, Government of India. "Schemes for Women Entrepreneurs." Available at: <https://wcd.nic.in>
- UN Women. "Women's Economic Empowerment." Available at: <https://www.unwomen.org/en/what-we-do/economic-empowerment>
- NITI Aayog, Government of India. "Women Entrepreneurship Platform (WEP)." Available at: <https://wep.gov.in>

13. Appendix

13.1. Sample Survey Questions (for Entrepreneur Feedback)

14. How did you first learn about government schemes for women entrepreneurs?
15. Have you ever abandoned a scheme application midway? If yes, what was the main reason?
16. Would you prefer a single online platform over visiting multiple department websites/offices?
17. How would you like to receive updates on your application or new schemes? (SMS / Email / App Notification)
18. Rate your overall satisfaction with the current process of accessing government support (1–5).

2. Additional Notes

This section may be used to attach wireframe screenshots prepared in Figma/Canva, additional diagrams, or sample generated reports/certificates once available.